CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 9 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

Q1. An electrically neutral object gains 5.0×10^{18} electrons. What will be the final net charge on the object? (a) $-0.80\ \mathrm{C}$ (b) $+0.80\ \mathrm{C}$ (c) $+4.80\ \mathrm{C}$ (d) $+0.20\ \mathrm{C}$ [PYQ 2021] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q2. Three point charges $+2q$, $-q$, and $-q$ lie perfectly at the three vertices of an equilateral triangle. The value of the electric field (E) and electric potential (V) strictly at the centroid of the triangle will be: (a) $E \neq 0$ and $V \neq 0$ (b) $E = 0$ and $V = 0$ (c) $E \neq 0$ and $V = 0$ (d) $E = 0$ and $V \neq 0$ [PYQ 2021] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q3. The specific resistance (resistivity) of a thick solid rod of pure copper as compared to that of a thin wire of pure copper at the exact same temperature is: (a) less (b) more (c) exactly the same (d) dependent upon the length of the wire [PYQ 2022] | **[Chapter 3: Current Electricity | Confidence: High]**

Q4. The time period of revolution of a charged particle undergoing uniform circular motion in a perpendicular magnetic field is strictly independent of its: (a) speed (b) mass (c) charge (d) magnetic field strength [PYQ 2020] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q5. An electron is released from rest in a region where a uniform electric field and a uniform magnetic field act strictly parallel to each other. The electron will: (a) move in a straight line (b) move in a circle (c) remain stationary (d) move in a helical path [PYQ 2020] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q6. A constant direct current is flowing through a long solenoid. An iron rod is now fully inserted into the solenoid along its central axis. Which of the following physical quantities will NOT increase? (a) The magnetic field at the centre (b) The magnetic flux linked with the solenoid (c) The rate of Joule heating in the wire (d) The self-inductance of the solenoid [PYQ 2021] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q7. The rms current in an AC circuit connected to a $50\ \mathrm{Hz}$ source is precisely $15\ \mathrm{A}$. The instantaneous value of the current exactly $1/600\ \mathrm{s}$ after the instant the current is zero and rising positively, is: (a) $15/\sqrt{2}\ \mathrm{A}$ (b) $15\sqrt{2}\ \mathrm{A}$ (c) $\sqrt{2}/15\ \mathrm{A}$ (d) $8\ \mathrm{A}$ [PYQ 2021] | **[Chapter 7: Alternating Current | Confidence: High]**

Q8. A welder typically wears special goggles or face masks with dark glass windows to protect his eyes mostly from the harmful effects of which electromagnetic waves produced by the welding arc? (a) Intense visible light (b) Infrared radiation (c) Ultraviolet rays (d) Microwaves [PYQ 2020] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q9. A small point object is placed at a distance of $10\ \mathrm{cm}$ in front of a concave mirror having a radius of curvature of $15\ \mathrm{cm}$. The exact position of the image formed is: (a) $+15\ \mathrm{cm}$ (b) $-30\ \mathrm{cm}$ (c) $-25\ \mathrm{cm}$ (d) $+5\ \mathrm{cm}$ [PYQ 2022] | **[Chapter 9: Ray Optics | Confidence: High]**

Q10. The geometrical shape of the interference fringes formed on the screen in a standard Young's double-slit experiment when the screen distance D is very large compared to the fringe width is practically: (a) straight lines (b) parabolic curves (c) concentric circles (d) hyperbolic curves [PYQ 2020] | **[Chapter 10: Wave Optics | Confidence: High]**

Q11. If incident photons of frequency ν fall on the surfaces of metals A and B having respective threshold frequencies of $\nu/2$ and $\nu/3$, the ratio of the maximum kinetic energy of electrons emitted from A to that from B is: (a) $2:3$ (b) $3:4$ (c) $1:3$ (d) $\sqrt{3}:\sqrt{2}$ [PYQ 2020] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Q12. A photon beam of wavelength $663\ \mathrm{nm}$ is incident on a metal surface whose work function is $1.50\ \mathrm{eV}$. The maximum kinetic energy of the emitted photoelectrons is roughly: (Take $h = 6.63 \times 10^{-34}\ \mathrm{J\,s}$, $1\ \mathrm{eV} = 1.6 \times 10^{-19}\ \mathrm{J}$) (a) $3.0 \times 10^{-20}\ \mathrm{J}$ (b) $6.0 \times 10^{-20}\ \mathrm{J}$ (c) $4.5 \times 10^{-20}\ \mathrm{J}$ (d) $9.0 \times 10^{-20}\ \mathrm{J}$ [PYQ 2023] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false.

Q13. Assertion (A): An electron has a physically higher potential energy when it is located at a point associated with a more negative value of electric potential. **Reason (R):** Electrons naturally tend to move from a region of higher electric potential to a region of lower electric potential. [PYQ 2023] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: High]**

Q14. Assertion (A): If a flexible conducting wire is bent into an irregular loop with its two ends fixed, and a steady current I is passed through it, the area enclosed by the loop strictly tends to increase, becoming circular. **Reason (R):** Adjacent segments of the wire carrying currents in strictly opposite directions strongly repel each other. [PYQ 2022] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q15. Assertion (A): A spherical convex mirror can never form a real image of any object under any optical circumstances. **Reason (R):** A convex mirror always diverges the parallel rays of light that are incident upon its reflecting surface. [PYQ 2020] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q16. Assertion (A): For incident radiation of a frequency greater than the threshold frequency, the saturation photoelectric current is strictly proportional to the intensity of the incident radiation. **Reason (R):** Greater the number of energy quanta (photons) available per second, greater is the number of electrons coming out of the metal per second. [PYQ 2023] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section B: 2-Mark Questions (Very Short Answer)

Q17. An ideal electric dipole is completely enclosed by a cubical Gaussian surface of side $1\ \mathrm{cm}$. State the net electric flux passing entirely through the cube, giving a valid scientific reason. [PYQ 2015] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q18. The relative magnetic permeability (μ_r) of a given material is exactly 0.5. Identify the specific magnetic nature of this material and formally write the mathematical relation linking relative permeability to its magnetic susceptibility (χ). [PYQ 2020] | **[Chapter 5: Magnetism & Matter | Confidence: High]**

Q19. Distinguish explicitly between the macroscopic electrical 'resistance' and the microscopic 'resistivity' of a metallic conductor on the basis of their dependence on physical dimensions. [PYQ 2020] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q20. A narrow slit is illuminated by a parallel beam of monochromatic light of wavelength $6000\ \backslash\mathrm{AA}$ and the angular width of the central maximum in the diffraction pattern is measured. When the slit is next illuminated by light of an unknown wavelength λ', the angular width strictly decreases by 30%. Calculate the exact value of the wavelength λ'. [PYQ 2022] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Q21. A proton and an α-particle are found to inherently possess the exact same de-Broglie wavelength. Determine the precise mathematical ratio of the accelerating potentials ($V_p : V_\alpha$) through which they must have been accelerated from rest. [PYQ 2015] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: High]**

Section C: 3-Mark Questions (Short Answer)

Q22. A total charge Q is distributed uniformly over a thin circular conducting ring of radius a. Obtain the mathematical expression for the electric field intensity E at a point on its central axis at a distance x from the centre. Hence, show logically that for points at large distances from the ring, it behaves strictly like a point charge. [PYQ 2016] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q23. Systematically derive the standard expression for the capacitance of a parallel plate capacitor having plate area A and plate separation d, with a vacuum (or air) present entirely between the two plates. [PYQ 2023] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q24. Two incandescent electric bulbs are specifically rated as (P_1, V) and (P_2, V). If they are connected (A) strictly in series, and (B) strictly in parallel across the exact same supply voltage V, deduce the formulas for the total power dissipated in the two combinations in terms of P_1 and P_2. [PYQ 2017] | **[Chapter 3: Current Electricity | Confidence: High]**

Q25. A straight metallic rod PQ of length l is rotated with a uniform angular velocity ω about an axis passing completely through its mid-point (O) and perpendicular to the plane of the rotation, in a uniform magnetic field B acting parallel to the axis of rotation. What is the net potential difference developed specifically between the two extreme ends of the rod, P and Q? Prove it mathematically. [PYQ 2020] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q26. An AC source generating a voltage $\varepsilon = \varepsilon_0 \sin(\omega t)$ is connected directly to a pure capacitor of capacitance C. Derive the formal expression for the instantaneous current I flowing through the circuit. Plot a graph showing the simultaneous variation of voltage ε and current I versus ωt to demonstrate their phase relationship. [PYQ 2022] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q27. A monochromatic ray of light is incident normally on the face AB of an equilateral triangular glass prism of refracting angle 60° made of a transparent material of refractive index $2/\sqrt{3}$. Trace the entire path of the ray as it passes through the prism, and calculate the exact angle of emergence (e) and the net angle of deviation (δ). [PYQ 2019] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q28. Briefly answer the following: (A) Illustrate, by giving a clear scientific example, how one can show that electromagnetic waves carry momentum and can exert radiation pressure. (B) How are microwaves typically produced? Why is it essential in a microwave oven to select the frequency of microwaves to exactly match the resonant frequency of water molecules? [PYQ 2014, 2019] | **[Chapter 8: Electromagnetic Waves | Confidence: High]**

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: Temperature Dependence of Resistivity The electrical resistance of a material fundamentally depends heavily on its temperature. In metallic conductors, an increase in temperature heightens the thermal agitation of lattice ions, which causes free electrons to collide much more frequently. Consequently, the relaxation time abruptly drops, increasing the resistivity. In contrast, for intrinsic semiconductors and insulators, an increase in thermal energy dramatically breaks covalent bonds, generating a massive exponential surge in the number density of free charge carriers (electrons and holes). This surge overwhelmingly compensates for any change in relaxation time. At exceptionally low temperatures near absolute zero, certain materials astonishingly exhibit zero electrical resistance, a highly specialized state called superconductivity. (A) Draw a qualitative graph showing the precise variation of resistivity (ρ) with absolute temperature (T) for a typical semiconductor. (1 Mark) (B) Give the physical reason why the resistivity of a semiconductor decreases with rising temperature despite an increase in collision frequency. (1 Mark) (C) The experimental plot between resistance (R) and temperature (T) for Mercury (Hg) shows its resistance suddenly dropping to exactly zero at $4.2\ \mathrm{K}$. What is this distinct phenomenon called, and what is the temperature $4.2\ \mathrm{K}$ called in this specific context? (2 Marks) [PYQ 2019 / Practice] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q30. Case Study: Diffraction at a Single Slit When a plane wavefront of monochromatic light is incident on a very narrow slit, it bends around the corners of the obstacle and continuously spreads into the geometric shadow region. This phenomenon is called diffraction. The resulting diffraction pattern produced on a distant screen consists of a broad, highly intense central bright maximum, symmetrically flanked by alternating, increasingly fainter secondary dark and bright fringes. The intensity of the secondary maxima strictly drops off rapidly. (A) If the physical width of the slit is decreased, what effect will it mathematically have on the angular width of the central maximum? (1 Mark) (B) Briefly explain why the maximum intensity of the first secondary maximum is significantly less than that of the central maximum. (1 Mark) (C) Two distinct wavelengths of sodium light, $590\ \mathrm{nm}$ and $596\ \mathrm{nm}$, are used in turn to study the diffraction taking place at a single slit of aperture $2 \times 10^{-6}\ \mathrm{m}$. If the distance between the slit and the observation screen is precisely $1.5\ \mathrm{m}$, calculate the linear separation on the screen between the positions of the second bright maxima of the diffraction patterns obtained in the two respective cases. (2 Marks) [PYQ 2019, 2022] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Section E: 5-Mark Questions (Long Answer)

Q31. (A) Draw the equipotential surfaces created purely by an ideal electric dipole. (B) In a parallel plate capacitor, the air between the plates has a dielectric constant of 1. Each plate has an area of exactly $6 \times 10^{-3}\ \mathrm{m^2}$ and the separation between the plates is strictly $3\ \mathrm{mm}$. (i) Calculate the intrinsic capacitance of this capacitor. (ii) If this capacitor is subsequently connected to a $100\ \mathrm{V}$ power supply, what will be the steady charge accumulated on each plate? (iii) How would the charge on the plates be quantitatively affected if a $3\ \mathrm{mm}$ thick mica sheet of dielectric constant $K = 6$ is entirely inserted between the plates while the voltage supply remains continuously connected? [PYQ 2022, 2020] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q32. (A) State the Biot-Savart Law, expressing the magnetic field due to a tiny current element purely in its vector form. (B) Apply the Biot-Savart Law to systematically derive the mathematical expression for the magnetic field intensity ($\vec{B}$) at a point situated on the central axis of a circular current-carrying loop of radius R, carrying current I, at a distance x from its geometrical centre. (C) Accurately draw the continuous magnetic field lines produced around and passing through a circular wire carrying a steady current. [PYQ 2015, 2016] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q33. (A) State Einstein's photoelectric equation, clearly explaining the physical significance of each symbol used. (B) Explain logically, utilizing Einstein's photoelectric equation, any two major observed experimental features in the photoelectric effect which cannot possibly be explained by the classical wave theory of light. (C) The threshold frequency of a specific metal is f. When monochromatic light of frequency $2f$ is incident on the metal plate, the maximum velocity of the emitted photoelectrons is v_1. When the frequency of the incident radiation is increased to $5f$, the maximum velocity of the photoelectrons is v_2. Find the exact mathematical ratio $v_1 : v_2$. [PYQ 2016, 2018] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Answer Key

Q1: (a) $-0.80\ \mathrm{C}$. (Wait, let's calculate: A neutral object gains 5.0×10^{18} electrons. Total charge $q = n(-e) = 5.0 \times 10^{18} \times (-1.6 \times 10^{-19}) = -0.80\ \mathrm{C}$. Note: Option (a) is correct. The object was neutral initially.) **Q2:** (c) $E \neq 0$ and $V = 0$. (Potential at centroid $V = k(2q - q - q)/r = 0$. The electric fields from the two $-q$ charges add up downwards, and the field from $+2q$ also points downwards. Thus, net $\vec{E}$ is strictly non-zero). **Q3:** (c) exactly the same. (Specific resistance, or resistivity, is an intrinsic property of the material and does not depend on physical dimensions like thickness or length). **Q4:** (a) speed. ($T = 2\pi m/qB$. The formula has absolutely no dependence on the velocity/speed v). **Q5:** (a) move in a straight line. (The electric field accelerates it linearly. Since $\vec{v}$ is parallel to $\vec{B}$, magnetic force $\vec{F}_m = -e(\vec{v} \times \vec{B}) = 0$. Thus, it strictly moves in a straight line). **Q6:** (c) The rate of Joule heating in the wire. (Inserting an iron core sharply increases permeability μ, raising the magnetic field, flux, and self-inductance. However, it does not change the Ohmic resistance of the wire or the DC current, so Joule heating I^2R remains constant). **Q7:** (a) $15/\sqrt{2}\ \mathrm{A}$. (Peak current $I_0 = 15\sqrt{2}\ \mathrm{A}$. $\omega = 2\pi(50) = 100\pi$. $I(t) = I_0 \sin(100\pi t) = 15\sqrt{2} \sin(100\pi \times \frac{1}{600}) = 15\sqrt{2} \sin(\frac{\pi}{6}) = 15\sqrt{2} \times 0.5 = 15/\sqrt{2}\ \mathrm{A}$). **Q8:** (c) Ultraviolet rays. (Welding arcs emit intense UV radiation which can cause severe retinal damage). **Q9:** (b) $-30\ \mathrm{cm}$. (

$R = -15$ cm $\implies f = -7.5$ cm. $u = -10$ cm.
$1/v + 1/u = 1/f \implies 1/v = 1/-7.5 - 1/-10 = -10/75 + 7.5/75 = -2.5/75 = -1/30$. Thus, $v = -30$ cm).
Q10: (a) straight lines. (At large distances, the inherently hyperbolic fringes have negligible curvature and appear as perfectly straight bands). **Q11:** (b) $3:4$. ($K_A = h\nu - h(\nu/2) = 0.5h\nu$. $K_B = h\nu - h(\nu/3) = (2/3)h\nu$. Ratio $K_A/K_B = 1/2 : 2/3 = 3/4$). **Q12:** (b) 6.0×10^{-20} J. (Incident Energy $E = hc/\lambda = (6.63 \times 10^{-34} \times 3 \times 10^{8})/(663 \times 10^{-9}) = 3 \times 10^{-19}$ J. Work function $W = 1.5$ eV $= 1.5 \times 1.6 \times 10^{-19} = 2.4 \times 10^{-19}$ J. $K_{max} = 3.0 \times 10^{-19} - 2.4 \times 10^{-19} = 0.6 \times 10^{-19}$ J $= 6.0 \times 10^{-20}$ J).

Q13: (c) A is true but R is false. (Electrons naturally accelerate from a region of lower potential to a region of higher potential, contradicting R). **Q14:** (a) Both A and R are true and R is the correct explanation of A. **Q15:** (d) A is false and R is also false. (A convex mirror *can* form a real image if the incident rays are highly converging (virtual object). Furthermore, while it diverges parallel rays, the absolute statement in A makes it false). **Q16:** (a) Both A and R are true and R is the correct explanation of A.

Q17: Exactly zero. A dipole contains equal and opposite charges ($+q$ and $-q$). Total enclosed charge is exactly 0. By Gauss's Law ($\Phi = Q_{enc}/\epsilon_0$), net flux is zero. **Q18:** Since $\mu_r < 1$, the material is diamagnetic. Relation: $\mu_r = 1 + \chi$. (Here, $\chi = -0.5$). **Q19:** Resistance directly depends on the physical dimensions (length and cross-sectional area) of the conductor. Resistivity is strictly an intrinsic material property and depends only on the nature of the material and its temperature, totally independent of its dimensions. **Q20:** Angular width $\theta = 2\lambda/a$. Since $\theta \propto \lambda$, if θ decreases by 30%, it becomes 0.7θ. Thus, new wavelength $\lambda' = 0.7 \times 6000 \text{\AA} = 4200 \text{\AA}$. **Q21:** $\lambda = h/\sqrt{2mqV}$. Since $\lambda_p = \lambda_\alpha$, $m_p q_p V_p = m_\alpha q_\alpha V_\alpha$. $m_p(e)V_p = (4m_p)(2e)V_\alpha \implies V_p = 8V_\alpha$. Ratio $V_p : V_\alpha = 8:1$.

Q22: Consider a small element dq on the ring. $dE = \frac{k(dq)}{r^2}$ where $r = \sqrt{x^2 + a^2}$. Resolve components: perpendicular components cancel, axial components $dE\cos\theta$ add up. $E = \int dE \frac{x}{\sqrt{x^2+a^2}} = \frac{kQx}{(x^2+a^2)^{3/2}}$. For $x \gg a$, a^2 is negligible. $E \approx \frac{kQx}{x^3} = \frac{kQ}{x^2}$, which precisely matches the field formula of a point charge. **Q23:** Let plates have charge $+Q, -Q$ and area A. Surface charge density $\sigma = Q/A$. Electric field between plates is uniform $E = \sigma/\epsilon_0 = Q/(A\epsilon_0)$. Potential difference $V = E \times d = Qd/(A\epsilon_0)$. Capacitance $C = Q/V = \frac{Q}{Qd/A\epsilon_0} = \frac{\epsilon_0 A}{d}$. **Q24:** Resistances are $R_1 = V^2/P_1$ and $R_2 = V^2/P_2$. (A) In series: $R_s = R_1 + R_2 = V^2/P_1 + V^2/P_2$. Power $P_s = V^2/R_s = \frac{V^2}{V^2/P_1 + V^2/P_2} = \frac{P_1P_2}{P_1+P_2}$. (B) In parallel: Equivalent resistance $1/R_p = 1/R_1 + 1/R_2$. Power $P_p = V^2/R_p = V^2(1/R_1 + 1/R_2) = P_1 + P_2$. **Q25:** The potential difference between the center (O) and one end (P) is $\frac{1}{2}B\omega(l/2)^2 = \frac{B\omega l^2}{8}$. The potential difference between the center (O) and the other end (Q) is also exactly $\frac{B\omega l^2}{8}$. Because both ends P and Q sweep identically and acquire the exact same polarity relative to the center, the net potential difference between P and Q is exactly ZERO ($V_P - V_Q = 0$). **Q26:** $q = C\varepsilon = C\varepsilon_0 \sin(\omega t)$. $I = dq/dt = \frac{d}{dt}(C\varepsilon_0 \sin\omega t) = C\varepsilon_0\omega\cos(\omega t) = I_0 \sin(\omega t + \pi/2)$. Plot a sine wave for voltage and a cosine wave for current starting at its positive peak, showing I leading ε by $90°$. **Q27:** Normal incidence means angle of incidence at AB is $i_1 = 0 \implies r_1 = 0$. Using $A = r_1 + r_2 \implies r_2 = 60°$. At face AC, apply Snell's Law: $\mu \sin(r_2) = 1 \times \sin e \implies (2/\sqrt{3})\sin 60° = \sin e \implies (2/\sqrt{3})(\sqrt{3}/2) = 1 \implies \sin e = 1 \implies e = 90°$ (Grazes the face AC). Angle of deviation $\delta = i_1 + e - A = 0° + 90° - 60° = 30°$. **Q28:** (A) A comet's tail always points strictly away from the sun due to the radiation pressure exerted by solar electromagnetic waves transferring momentum to the dust particles. (B) Produced by vacuum tube devices like Klystrons/Magnetrons. The microwave frequency must match the resonant frequency of water to maximize energy transfer via resonance, intensely vibrating the molecules to efficiently heat the food.

Q29: (A) Graph of resistivity (ρ) vs temperature (T) showing a smooth, exponentially decaying curve starting high on the y-axis and approaching zero at high T. (B) In semiconductors, rising temperatures break covalent bonds, causing an exponential increase in charge carrier density (n) which vastly outweighs the slight decrease in relaxation time, dropping overall resistivity. (C) This absolute loss of resistance is called Superconductivity. The temperature 4.2 K is called its 'Critical Temperature' or 'Transition Temperature' (T_c). **Q30:** (A) Angular width of central max $= 2\lambda/a$. If slit width 'a' decreases, the angular width strictly increases. (B) For the 1st secondary maximum, only one-third of the exposed wavefront remains uncancelled by destructive interference, yielding a drastically lower amplitude and intensity compared to the central maximum. (C) Position of 2nd bright fringe $y_2 = \frac{5\lambda D}{2a}$. Separation $\Delta y = \frac{5D}{2a}(\lambda_2 - \lambda_1) = \frac{5 \times 1.5}{2 \times (2 \times 10^{-6})}(596 - 590) \times 10^{-9} = \frac{7.5}{4 \times 10^{-6}} \times 6 \times 10^{-9} = 1.875 \times 6 \times 10^{-3} = 11.25 \times 10^{-3}$ m $= 11.2$
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Q31: (A) Draw two equal and opposite point charges. Equipotential surfaces are eccentric spheres around each charge (closer together between the charges) and a flat straight vertical plane precisely bisecting the line joining the two charges. (B) (i) $C_0 = \epsilon_0 A/d = (8.85 \times 10^{-12} \times 6 \times 10^{-3})/(3 \times 10^{-3}) = 17.7 \times 10^{-12}$ F $= 17.7$ pF. (ii) $Q_0 = C_0 V = 17.7 \times 10^{-12} \times 100 = 1.77 \times 10^{-9}$ C $= 1.77$ nC. (iii) With battery connected, V remains exactly 100 V. $C_{new} = KC_0 = 6 \times 17.7$ pF $= 106.2$ pF. New charge $Q_{new} = C_{new}V = 106.2 \times 10^{-12} \times 100 = 10.62 \times 10^{-9}$ C $= 10.62$ nC (Charge strictly increases by a factor of 6). **Q32:** (A) $d\vec{B} = \frac{\mu_0 I}{4\pi}\frac{d\vec{l} \times \vec{r}}{r^3}$. (B) Let point P be at distance x on the axis. Consider top element dl. $r = \sqrt{R^2 + x^2}$. $dB = \frac{\mu_0 I dl}{4\pi(R^2+x^2)}$. Resolving dB, vertical components strictly cancel out by symmetry. Axial components add up: $dB_x = dB\sin\theta = dB\frac{R}{\sqrt{R^2+x^2}}$. Total field $B = \int dB_x = \frac{\mu_0 IR}{4\pi(R^2+x^2)^{3/2}}\oint dl$. Since $\oint dl = 2\pi R$, $B = \frac{\mu_0 IR^2}{2(R^2+x^2)^{3/2}}$. (C) Draw a cross-section of the loop. Field lines form closed loops around the wire segments, strictly pointing straight along the geometric axis in the centre. **Q33:** (A) $K_{max} = h\nu - \Phi$. (K_{max} is max kinetic energy of photoelectron, h is Planck's constant, ν is frequency of incident light, Φ is the inherent work function of the metal). (B) 1. Wave theory falsely predicts that higher intensity should automatically yield higher kinetic energy; Einstein showed KE depends solely on frequency. 2. Wave theory cannot explain the existence of a sharp threshold frequency below which no emission occurs, regardless of high intensity; Einstein showed photons must individually have enough minimum energy ($h\nu_0$) to eject an electron. (C) By Einstein's equation: $K_{max} = \frac{1}{2}mv^2 = h\nu - hf$. For $\nu = 2f$: $v_1^2 \propto h(2f) - hf = hf$. For $\nu = 5f$: $v_2^2 \propto h(5f) - hf = 4hf$. Ratio $v_1^2/v_2^2 = hf/4hf = 1/4 \implies v_1 : v_2 = 1 : 2$.

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